Note 7 Moving-Average Models

1 Moving-average (MA) Models

If a time series $\{X_t\}$ is generated from the following model

$$X_t = \mu + W_t - \theta_1 W_{t-1} - \theta_2 W_{t-2} - \cdots - \theta_q W_{t-q},$$

where $\{W_t\}$ is white noise with mean zero and variance σ_W^2, then $\{X_t\}$ is called a **moving-average model of order q**, denoted as MA(q).

Remarks:

1. The MA process is a linear process. Hence a MA process is always **stationary**.

2. It is usually convenient to assume normality of the white noise, that is,

   $$X_t = \mu + Z_t - \theta_1 Z_{t-1} - \theta_2 Z_{t-2} - \cdots - \theta_q Z_{t-q},$$

   where $\{Z_t\}$ is Gaussian white noise with mean 0 and variance σ_Z^2. This assumption can be quite helpful in parameter estimation and forecasting.

3. $\mathbb{E}(X_t) = \mu$

4. Specifically, a MA(1) process is

$$X_t = \mu + Z_t - \theta_1 Z_{t-1}$$

A MA(2) process is

$$X_t = \mu + Z_t - \theta_1 Z_{t-1} - \theta_2 Z_{t-2}$$

5. Using the backshift operator, an MA(q) model can be written as

2 Invertibility

If a time series $\{X_t\}$ can be expressed in the following form

$$X_t - \sum_{i=1}^{\infty} \pi_i X_{t-i} = C + Z_t,$$

with $\sum_{i=1}^{\infty} |\pi_i| < \infty$, then $\{X_t\}$ is said to be invertible.

Why do we need to consider invertibility of a MA process?

Generally, a MA(q) process is invertible if and only if all the roots of

$$\theta(B) = 1 - \theta_1 B - \cdots - \theta_q B^q = 0$$

are outside the unit circle.

Example 2.1. Consider the following two MA processes, are they invertible?

- $X_t = 5 + Z_t + 0.1Z_{t-1} - 0.3Z_{t-2}$
- $X_t = 8 + Z_t + 0.5Z_{t-1} - 0.5Z_{t-2}$

2.1 π weights

If a MA process is invertible, how can be find its invertible form? That is, how can we know the values of C and π_is?

Example 2.2. Consider the invertible MA process below

$$X_t = 5 + Z_t + 0.1Z_{t-1} - 0.3Z_{t-2}$$

Find the π weights.

3 Model Identification

In practice, if we are given a series. How can we know a MA process is appropriate to model the series? And how can we choose its order q?

Generally, we need to consider the ACF and PACF.

3.1 Autocorrelation Functions

- ACF of MA(1)

- ACF of MA(2)

- ACF of MA(q)

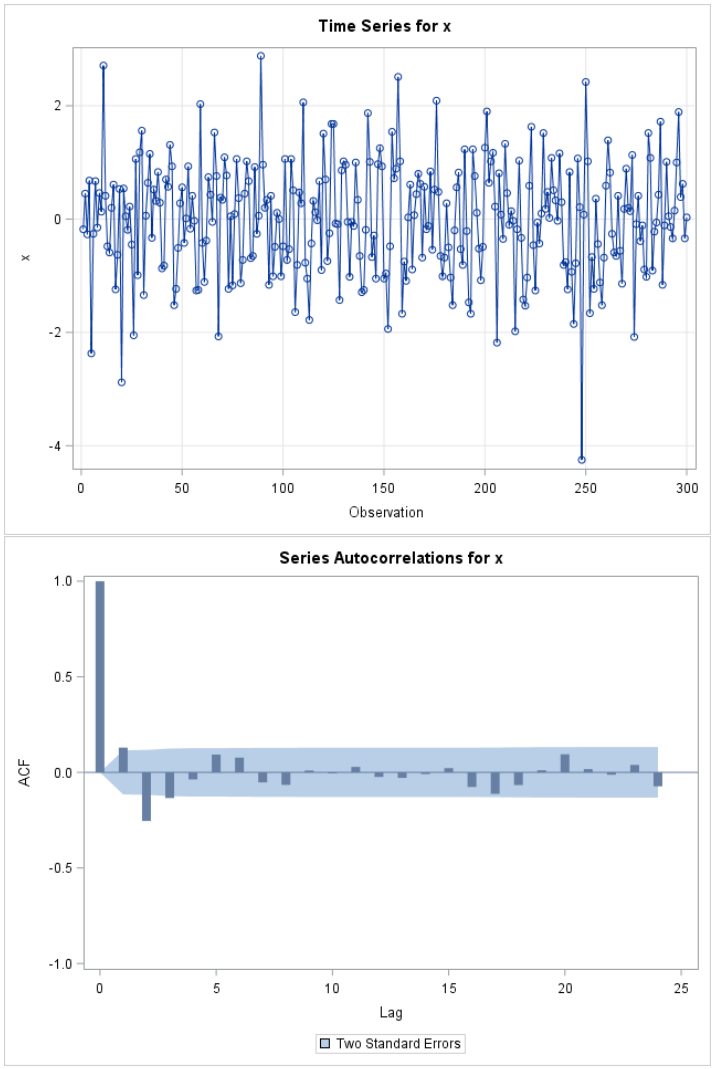
Generally, a MA(q) process has the following autocorrelation function

$$\rho(h) = \begin{cases} -\frac{\theta_h + \theta_1\theta_{h+1} + \cdots + \theta_{q-h}\theta_q}{1+\theta_1^2+\cdots+\theta_q^2}, & 1 \le h \le q; \\ 0 & h > q \end{cases}$$

Note: the ACF of the MA(q) **CUTS OFF** at q.

3.2 How to determine q?

Below is the plot of a series $\{X_t\}$ with $n = 300$. Which model we should use to fit the series?



4 Estimation

For a MA(q) process, what parameters we have to estimate?

Common estimation methods for MA processes are

- Maximum likelihood method
- Unconditional least square method
- Conditional least square method

4.1 Maximum Likelihood Method

Let $\{X_t\})_{t=1}^n$ be a time series of length n generated from a MA(q) model,

$$X_t = \mu + Z_t - \theta_1 Z_{t-1} - \theta_2 Z_{t-2} - \cdots - \theta_q Z_{t-q},$$

with $\{Z_t\} \sim WN(0, \sigma_Z^2)$.
Since the $\{Z_t\}$ is Gaussian white noise, then $\{X_t\}$ becomes a multivariate normal random variable with mean vector $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Gamma}$, where

$$\boldsymbol{\Gamma} = \begin{bmatrix} \mathrm{Cov}(X_1, X_1) & \mathrm{Cov}(X_1, X_2) & \cdots & \mathrm{Cov}(X_1, X_n) \\ \mathrm{Cov}(X_2, X_1) & \mathrm{Cov}(X_2, X_2) & \cdots & \mathrm{Cov}(X_2, X_n) \\ \vdots & \vdots & \ddots & \vdots \\ \mathrm{Cov}(X_n, X_1) & \mathrm{Cov}(X_n, X_2) & \cdots & \mathrm{Cov}(X_n, X_n) \end{bmatrix} = \begin{bmatrix} \gamma(0) & \gamma(1) & \cdots & \gamma(n-1) \\ \gamma(1) & \gamma(0) & \cdots & \gamma(n-2) \\ \vdots & \vdots & \ddots & \vdots \\ \gamma(n-1) & \gamma(n-2) & \cdots & \gamma(0) \end{bmatrix}$$

The likelihood function of $\{X_t\}$ becomes

$$L(\mu, \theta_1, \cdots, \theta_q, \sigma_Z^2 | X_1, \cdots, X_n) = \frac{1}{\sqrt{2\pi|\boldsymbol{\Gamma}|}} \exp\left(\frac{1}{2}(\boldsymbol{x} - \boldsymbol{\mu})' \boldsymbol{\Gamma}^{-1} (\boldsymbol{x} - \boldsymbol{\mu})\right)$$

Denote $\boldsymbol{\eta} = (\mu, \theta_1, \theta_2, \cdots, \theta_q, \sigma_Z^2)$. The likelihood function can be rewritten as

$$L(\boldsymbol{\eta} | X_1, \cdots, X_n)$$

We obtain the estimate $\hat{\boldsymbol{\eta}} = (\hat{\mu}, \hat{\theta}_1, \hat{\theta}_2, \cdots, \hat{\theta}_q, \hat{\sigma}_Z^2)$ by maximizing the L function.

$$\hat{\boldsymbol{\eta}} = \mathrm{argmax}\, L(\boldsymbol{\eta} | X_1, \cdots, X_n)$$

Example 4.1. Find the maximum likelihood estimates for the $\{X_t\}$ series on page 11 if a MA(2) process is used to model the series.

What about a MA(3) process?

Example 4.2. In the previous example, should we use a MA(2) or MA(3) to fit the series?

4.2 Unconditional Least Square Method

The unconditional least square (ULS) method is an alternation of the MLE that is computationally easier in the optimization step. Let $\boldsymbol{X} = (X_1, \cdots, X_n)'$ and

$$S(\boldsymbol{\eta}|\boldsymbol{X}) = \boldsymbol{X'\Gamma^{-1}X},$$

then maximizing the likelihood function is approximately equivalent to minimizing $S(\boldsymbol{\eta}|\boldsymbol{X})$.
The unconditional least square estimates are

$$\hat{\boldsymbol{\eta}} = \operatorname{argmin} S(\boldsymbol{\eta}|\boldsymbol{X})$$

The unconditional least squares estimates are obtained by minimizing the sum of squared residuals.

4.3 Conditional Least Square Method

The CLS estimates are conditional on the assumption that the past unobserved errors are equal to 0. Recall if a MA process $\{X_t\}$ is invertible, it can be expressed in the following form

$$X_t - \sum_{i=1}^{\infty} \pi_i X_{t-i} = C + Z_t.$$

The conditional least square method obtains the estimates by minimizing

$$S(\boldsymbol{\eta}|\boldsymbol{X}) = \sum_{t=1}^{n} Z_t^2 = \sum_{t=1}^{n} (X_t - \sum_{i=1}^{\infty} \pi_i X_{t-i})^2,$$

where the unobserved past values of X_t are set to 0 and the π weights can be computed by methods we mentioned before.

4.4 Model Selection

5 Diagnostics

If we identified the order q of a MA(q) process and estimated the parameters correctly, the residuals should be approximately white noise and satisfy the main assumptions of the model. For a MA(q) process, the residuals are

$$r_t = X_t - \hat{X}_t, \quad t = 1, \cdots, n$$

where $\hat{X}_t$ is the fitted value at time t. Computation of $\hat{X}_t$ will be introduced later.

Two assumptions are usually checked for the residuals:

- independence;
- normality.

5.1 Independence Check

5.1.1 Ljung-Box Test

5.1.2 Individual Check

5.2 Normality Check

6 Forecasting

6.1 Point Estimate

Suppose we have correctly identified a MA(q) process and the parameters are well estimated, say

$$X_t = \hat{\mu} + Z_t - \hat{\theta}_1 Z_{t-1} - \hat{\theta}_2 Z_{t-2} - \cdots - \hat{\theta}_q Z_{t-q}.$$

We cannot use this model to forecast a future X_{t+m} since the past errors Z_t's are not observed.

Recall: A MA(q) model is invertible if it can be expressed in the following form

$$X_t - \sum_{i=1}^{\infty} \pi_i X_{t-i} = C + Z_t,$$

with $\sum_{i=1}^{\infty} |\pi_i| < \infty$. As we have mentioned, we can only use a MA(q) model to forecast when it is **invertible**.

- Let $\hat{X}_{t+1}$ be the 1-step predictor of a invertible MA process, then

$$\hat{X}_{t+1} = \hat{C} + \hat{\pi}_1 X_{t-1} + \cdots + \hat{\pi}_{t-1} X_1.$$

- The m-step predictor is

$$\hat{X}_{t+m} = \hat{C} + \hat{\pi}_1 X_{t+m-1} + \cdots + \hat{\pi}_{t+m-1} X_1.$$

  If any X_t is unobserved, replace it by $\hat{X}_t$. Here, $\hat{\pi}_i$ are functions of $\hat{\phi}_i$.

6.2 Prediction Interval